

Editorial

Advancing the science of visualization of health data for lay audiences

Introduction

In this issue, we focus on the timely need to communicate best practices and practical, robust applications of designing and evaluating health data visualizations for lay audiences. We define lay audiences as those interacting with informatics tools in a non-professional capacity (eg, patients, caregivers, community members, research participants), as they may have distinct needs from health professionals. Since the Health Information Technology for Economic and Clinical Health (HITECH) Act incentivized the utilization of clinical informatics systems, the volume of health data that learning health systems are collecting and aggregating on patients has grown exponentially. Patients are also generating their own data through digital health tools that they and the health system want to leverage to improve health. In parallel with vast quantities of data, the 21st Century Cures Act requires that electronic health information be freely accessible and authorizes penalties for those who block data from patients. There are also non-clinical streams of data (eg, environmental exposure, disease transmission) that are increasingly accessible to the public. Though barriers to accessing data are being lifted, the data are often available in a raw format that is rarely comprehensible without a significant amount of preprocessing. Once processed, data may still require contextualization to the person or the community of interest to make it actionable. Therefore, the development and evaluation of visualizations of health data for lay audiences is an important area of inquiry.

Despite well-intentioned investment, there are substantive gaps in using rigorous methodology and appropriate human-centered design to support the development of meaningful health visualizations for patients, family caregivers, and the public. A systematic review of patient-facing visualizations demonstrated that almost half of the included studies did not involve relevant lay audience end-users at all in the design or evaluation process, and there was a lack of consistency and rigor in the methods described for those that did involve lay audiences.¹ Moreover, others that involved lay audience end-users had fundamental misconceptions regarding human-centered design, such as assuming that soliciting user requests was sufficient.² Studies that do involve lay audiences also often fail to go beyond an initial assessment of user preferences, leading to the development of interventions that are not actionable, understandable, or do not lead to the desired outcomes.² When studies do involve lay audience participants in visualization development, participants are often not representative of the diversity of the intended audience's

characteristics such as level of education, race/ethnicity, age, health status, literacy/numeracy, experience with technology, and physical abilities.³

The onset of the COVID-19 pandemic spurred innovation in the development and accelerated adoption of remote tools for clinical care (ie, telehealth) and research. For example, many participatory design sessions moved online leading to the development of innovative mechanisms for obtaining user feedback that mitigate geographic barriers. There is an opportunity to blend remote, hybrid, and in-person approaches to gain inclusive perspectives for human-centered design of health data visualizations. Designing for lay audiences is inherently challenging because their needs are often not well-articulated, and their requests may even violate basic design principles. Methods that work well for domain experts (eg, health professionals) may not be appropriate for lay audiences.

Given these challenges, there is a need to understand effective approaches and present practical applications for designing and evaluating visualizations for lay audiences. This focus issue includes 14 papers selected from among 24 submissions through rigorous peer review. The works present overarching perspectives, specific applications of visualization design, tutorials, and reviews to advance the science. Three of the papers have authorship teams that are entirely new to JAMIA.

The collection of papers in this issue adds to efforts from others supporting human-centered design of data visualization. The American Medical Informatics Association (AMIA), for example, has a working group on visual analytics that promotes the development and validation of new visual analytical techniques that can be used to address some of the fundamental data and informatics challenges evident in the healthcare domain. In contrast to this focus issue, the Visual Analytics Working Group (VIS-WG) does not focus solely on lay audiences. The updated International Patient Decision Aid Standards provide helpful best practices for how and when visual formats should be used,⁴ but do not address the issue highlighted in Ancker et al.⁵ regarding what may be the best visualization format based on the goal of the visualization designer. The Data Visualization Society, though not explicitly focused on health-related data, provides resources to members to support the growth, refinement, and expansion of data visualizations through mechanisms like their Slack channel, *Nightingale* magazine, and their annual "Information is Beautiful" awards. Governmental organizations have also recognized the need to support communication of health data to lay audiences; the National Library of Medicine recently issued a funding call to support, "Personal

Health Informatics for Delivering Actionable Insights to Individuals.” This announcement is also supported by the National Institute of Mental Health.

These efforts in addition to the focus issue underscore the important role visualizations will continue to play in communicating health-related information to lay audiences. In this editorial, we summarize the contributions of the focus issue to the state of the science of data visualization for lay audiences. As with any collection of articles, the included papers also reveal gaps in current literature and crucial next steps for advancing the field.

Overview

This focus issue highlights scholarly work representing the state of the art in visualization of health data for lay audiences. In [Table 1](#), the papers are summarized according to the type of contribution they make, their objective, and the intended visualization audience.

Among the papers in this focus issue are five design studies describing the refinement of visualizations delivered via public health dashboards (sexually transmitted infections),⁶ mobile apps (smoking cessation, sleep tracking),^{7,18} and patient education materials (genomic results, prostate cancer trial outcomes).^{8,16} Also included are two studies that used experimental designs to compare the performance of various presentation formats (text, number lines)¹² and visualization features (interactivity, tailoring, explanatory text)¹⁰ on outcomes such as classification accuracy, recall, preferences, perceived risk, behavioral intentions, and trust. Notably, only one study—a pilot among patients living with HIV—examined the effect of finalized patient education visualizations on clinical outcomes including disease-related knowledge, biomarkers, and patient-reported outcomes.¹⁷ Importantly, the above studies relied heavily on members of the target lay audience to ensure the relevance and appropriateness of the resulting visualizations. One scoping review, however, challenges us to move beyond including lay people only as research participants, proposing instead that there is a place for community engagement at every stage of the visualization development process.⁹ A perspective piece further challenges us to improve the validity and comparability of our findings by defining our visualization goals and outcomes in consistent, distinct, and granular ways.⁵ Continuing innovations in our methods, teams, and tools will be needed to help us achieve our visualization goals. One perspective adds immersive virtual reality (IVR) to our suite of visualization research methods¹³ whereas another seeks to foster effective teams by bridging the knowledge and expectation gaps that often exist between researchers and design professionals.¹⁵ Lastly, two papers describe tools—style guides¹⁴ and a template-based design tool¹¹—that facilitate the creation of consistently high-quality infographics.

Highlights

In this section, we highlight five papers that focus on tools and methods because they will be of general interest to readers and can help us advance the science of visualization. Conceptual clarity and methodological rigor are key to scientific advancement. In one of our Editor’s Choice papers, Ancker et al.⁵ point out that studies purporting to evaluate “comprehension” or “understanding” are, in fact, examining

a broad range of outcomes. For example, recalling a numerical value is a distinctly different task from correctly categorizing that value within given reference ranges; a visualization format that facilitates the former may or may not support the latter. Inadequate conceptual clarity around outcomes is a problem because study results cannot be meaningfully compared when they are measuring different things, nor can we come to valid conclusions about what visualization formats are best if we have not answered the question, “best for what goal?” To address this lack of standardization, the authors propose a taxonomy of “14 distinct and mutually exclusive cognitive, perceptual, and behavior outcomes” drawn from the visualization literature.⁵ Using this taxonomy can help visualization designers tackle the critically important step of clearly defining their communication goals^{19,20} and subsequently selecting appropriate, granular outcomes.

In our other Editor’s Choice paper, we learn from Mangal et al.¹⁵ that the ability to clearly define project goals is also essential for forging collaborative partnerships between researchers and design professionals, such as graphic designers, illustrators, and user interface/user experience designers. Drawing upon their experiences collaborating with each other, the authors provide a practical guide to avoiding common pitfalls before, during, and after the design process. Researchers—who may be unaware of the needs of design professionals—are likely to benefit from the glossary of key terms in the paper and the scope of work and design brief templates in the supplemental materials. The design brief, in particular, can help teams to crystallize their visualization goals, stakeholders, mandatory specifications, assets, deliverables, and more into one summary document. Given the readership of the journal, this paper is targeted primarily to researchers. However, it can also provide valuable insights to design professionals who may be unfamiliar with how the needs of academics differ from those of more typical corporate clients.

One approach to visualization that can be borrowed from the corporate world is the use of data visualization style guides. Much in the same way that organizations issue brand identity guidelines to standardize fonts, colors, logos, etc., so too can they provide detailed guidance for data and information visualizations produced by and for the organization. Graze and Schwabish¹⁴ offer a practical tutorial defining the recommended components of a data visualization style guide with a link to numerous examples. In the second part of the paper, they focus on the color palette section of the style guide, including instructions for implementation in widely used software packages. Style guides set a quality standard for visualizations and reduce the decision-making burden visualization designers face.

Visualization templates provide even more structure to support non-expert visualization designers. Cullen et al. describe the development and usability testing of a template-based public health-specific infographic design tool, Florence.¹¹ This tool improves upon existing commercially available software by offering health-related visual elements as well as bi-directional graph-text binding wherein changes to the data are synced across both graphical and textual representations of those data. The features of Florence and the simplicity of its interface are designed to meet the specific needs of public health professionals who must create visualizations for the public under tight time and budgetary constraints.

Table 1. Summary of articles by type of contribution, paper objective, and intended visualization audience.

First author	Type of contribution	Paper objective	Intended visualization audience
Ancker ⁵	Methodological critique	To propose a taxonomy of cognitive, perceptual, and behavioral outcomes useful for granular evaluation of the extent to which information visualizations achieve their communicative objectives	Any
Ansari ⁶	Visualization refinement	To develop a public-facing state-level dashboard displaying data about sexually transmitted infections	Public health professionals and the general public
Brin ⁷	Visualization refinement	To refine a smoking cessation mobile app for people living with HIV through heuristic evaluation with experts and usability testing with end-users	People living with HIV
Casillan ⁸	Visualization refinement	To develop educational materials about genome-informed risk assessments	Patients receiving genomic results
Chau ⁹	Scoping review	To (1) synthesize literature on the types of community engagement used in development of data visualizations intended to improve population health and (2) characterize instances in which data visualizations developed in community-researcher partnerships demonstrate “creative data literacy”	Community members
Cotter ¹⁰	Visualization format testing	To test the influence of interactivity and descriptive titles on dashboard users’ perceived susceptibility to the flu, intention to vaccinate, and information recall	General public, with focus on older adults
Cullen ¹¹	Tool development	To create and evaluate a template-based infographic design webtool for use by public health practitioners	Any
Desai ¹²	Visualization format testing	To compare four formats for conveying machine learning-derived (ML) postpartum depression (PPD) risks impact on: patient classification of recommended actions (primary outcome) and intention to seek care, perceived risk, trust, and preferences (secondary outcomes)	Patients with potential risk of postpartum depression
Ferguson ¹³	Methodological innovation	To introduce readers to the opportunities afforded by immersive virtual reality to study user-initiated information search among health information visualizations and to highlight critical study design considerations when using immersive virtual reality for this purpose	Research participants
Graze ¹⁴	Tutorial	To (1) promote the uptake of visualization style guides by highlighting their value and describing their key components and (2) provide detailed guidance on the section about color palettes	Any
Mangal ¹⁵	Tutorial	To facilitate successful collaborations between health researchers and design professionals by providing practical guidance on averting common pitfalls	Any
Ruzich ¹⁶	Visualization refinement	To develop a brief, informative, and understandable resource sheet summarizing patient-reported outcome (PRO) data from a prostate cancer clinical trial concordant with patients’ information needs and preferences	Patients with prostate cancer
Stonbraker ¹⁷	Clinical outcomes	To pilot test the effects of an infographic-based health communication intervention for Latino persons with HIV on HIV-related knowledge, self-efficacy to manage HIV, adherence to antiretroviral therapy, CD4 count, viral load, and current and overall health status	People living with HIV
Tsai ¹⁸	Visualization refinement	To (1) explore user experiences with Fitbit app sleep data visualizations and (2) identify areas for improvement and accompanying potential solutions	Wearable device users

A perspective by Ferguson et al. suggests an innovative way researchers can overcome practical constraints to studying “how lay people seek, acquire, and interpret health information in everyday settings.”¹³ A team at the Advanced Visualization Branch of the National Institute of Nursing Research developed an IVR grocery store that allows users to select products and view and compare their nutrition labels. This approach to studying behaviors relevant to health self-management is attractive because participants can interact with visualizations “in the wild” without the need for creating resource-intensive physical simulation environments or subjecting participants to potential safety issues in uncontrolled real-world environments. Readers interested in using

IVR in similar ways will benefit from the authors’ discussion of critical study design considerations.

Conclusions and future directions

We, the guest associate editors, foresee that in the coming years, information visualization will become an ever more important part of informatics, whether we are communicating with lay audiences, with professionals in other fields, or among ourselves. The growing prominence of visualization will be accompanied by rising expectations about the sophistication and quality of the visualizations we produce. This forecast has implications for our collaborations, our training,

the tools we use, the resources we share, and our approach to accessibility.

One lesson to be drawn from this focus issue is the value of bringing in a range of perspectives throughout the visualization process, especially those of design professionals and members of the target audience. In practice, this means (1) doing community engagement work and forming collaborative teams at the earliest stages of a project, (2) drafting budgets that reflect the value we place on supporting varied perspectives, and (3) transparently documenting our community engagement and collaboration practices in our published reports.⁹ Visualization is rapidly shifting from being a non-essential “nice to have” to being a “must have.” As such, adequate compensation and resources must be set aside to support collaborators’ contributions. Organizations large enough to support multiple projects may find it feasible and desirable to keep design professionals on staff as a shared resource rather than sending work out to freelancers.

Although people working in informatics come from a variety of professions, we observe that one thing many of us have in common is little to no formal training in even the most essential visualization skills, such as selecting an appropriate chart based on the type of data to be displayed and the goal of the visualization designer. Furthermore, because we do not know what we do not know, we are at risk of overestimating our abilities as visualization researchers and designers.²¹ As such, our desire to keep pace with increasing visualization demands must be matched by investments in training, even in clinical programs. We have learned from Ancker et al.⁵ and Mangal et al.¹⁵ the critical importance of crafting a specific and measurable visualization objective. Apart from that, it is not yet clear which other competencies should be considered foundational for all and which will be needed only by people more actively engaged in various kinds of visualization work. Therefore, a flexible, multidisciplinary framework of visualization competencies would be of substantial value for curriculum design and represents a promising area of scholarly inquiry.

One of the factors driving high expectations about visualization quality is the availability of tools such as Tableau, PowerBI, and Flourish. Despite these resources, additional tools will be needed to support visualization creation. In particular, we need more tools like Florence¹¹ that democratize the creation of attractive, high-quality visualizations by people with varying skill levels. Generative artificial intelligence (AI), such as OpenAI’s image model, DALL-E 3,²² may also provide resources for visualization development. However, as with language-based generative AI, concerns have been raised about the provenance of the training data and the fact that these models may simply perpetuate existing stereotypes, biases, and poor existing visualization designs.^{23,24} While generative AI holds potential for improving the efficiency of visualization design, rigorous scrutiny of the quality and inclusiveness of the designs created is sorely needed. Organizations like universities and health care systems would be wise to invest in creating visualization style guides and (perhaps more importantly) training their affiliates in how to use expert-made guides, as well as safe, effective collaboration with AI-based tools. The expected return on that investment is less time and effort for employees and better, more consistent visualizations.¹⁴

Despite efforts to share best practices and lessons learned in venues such as JAMIA and AMIA’s aforementioned VIS-

WG, visualization projects are at risk of expending time and resources reinventing the wheel and rediscovering what prior projects have already learned. To support scalable and efficient growth, and in the spirit of open science, we need centralized repositories of resources such as graphical components,¹¹ templates, boilerplate language, and finalized infographics that are available for adaptation and reuse. Ideally, these resources would be accompanied by supporting documentation, such as the evidence underlying design decisions, the populations/contexts with which the resources were developed, implementation guidance, technical documentation for code-based assets, permissions/licenses required, and any associated publications.

Greater use of—and indeed reliance on—visualizations increases our obligation to address the many facets of accessibility. This means moving beyond simply asking ourselves if font sizes are big enough and colors are distinguishable to people with color vision deficiencies. Making visualizations more accessible also means using inclusive, non-stigmatizing language,²⁵ offering visualizations in the viewer’s language of choice, providing visualization-creation tools for non-experts, evaluating the impact of interactivity on photosensitive epilepsy,²⁶ designing for lower levels of health and graph literacy, and providing useful, usable alt-text or other options²⁷ for those who may not be able to access visualizations.³ By improving accessibility, visualizations will be better able to help lay audiences make sense of health information, manage their health conditions, and engage in shared decision making.

Visualizations are clearly an increasingly important vehicle for communicating health-related data to lay audiences. This focus issue revealed that visualization-based communications for lay audiences are still in development stages based on the number of papers simply comparing visualization formats, as opposed to other domains of informatics, which have moved on to focus on the effects of interventions on health outcomes. To move past these foundational stages and iterations that continue to reinvent the wheel, we need to synthesize literature on the best visualization format for a given visualization goal. As described by Ancker et al.⁵, the ability to do this requires more standardized taxonomies for measuring visualization-related outcomes. The state of the science will further be supported by development of core competencies for visualization designers, developing new tools to support efficient visualization design while safely leveraging the advantages of AI, and creating open-source repositories for sharing reproducible visualizations. This must all be done while embracing a philosophy of human-centered, universal design to ensure visualizations are accessible by diverse groups of lay audiences.

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